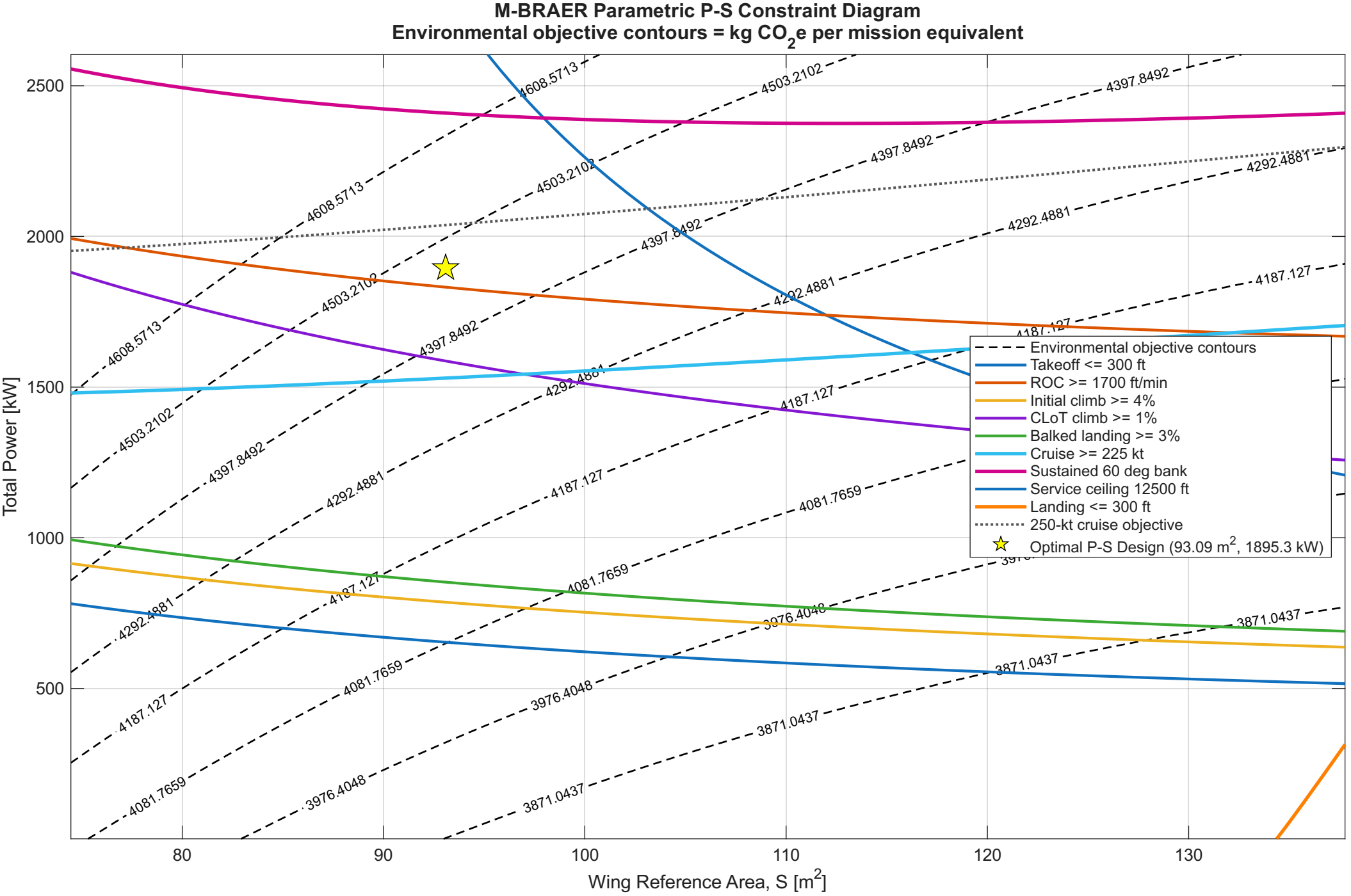
M-BRAER Parametric P-S Constraint Diagram
Environmental objective contours = kg CO2e per mission equivalent
Total Power [kW]
Wing Reference Area, S [m2]
2500
2000
1500
1000
500
80
90
100
110
120
130
4608.5713
4503.2102
4397.8492
4292.4881
4187.127
4081.7659
3976.4048
3871.0437
Environmental objective contours
Takeoff <= 300 ft
ROC >= 1700 ft/min
Initial climb >= 4%
CLoT climb >= 1%
Balked landing >= 3%
Cruise >= 225 kt
Sustained 60 deg bank
Service ceiling 12500 ft
Landing <= 300 ft
250-kt cruise objective
Optimal P-S Design (93.09 m2, 1895.3 kW)